

| Question | Marking Guidance                                                                                                                                                                                                                              | Mark                      | Comments                                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| 07.1     | <p>1. (Only) the presynaptic neurone/knob/membrane releases/has neurotransmitter/acetylcholine;</p> <p>2.(Only) the postsynaptic neurone/membrane has receptors</p> <p><b>OR</b></p> <p>No receptors in the presynaptic neurone/membrane;</p> | <p>2</p> <p>(2 x AO1)</p> | <p>1. Accept abbreviations for acetylcholine e.g., ACh, Ach, AChol.</p> <p>1. Ignore has/releases 'transmitter'.</p> |

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                        | Mark                      | Comments                                                                                                                                                                                                                                  |
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| 07.2     | <p>1. Temporal;</p> <p>2. (Several/repeated impulses in short time) provide (enough) neurotransmitter/acetylcholine to reach threshold</p> <p><b>OR</b></p> <p>(Several/repeated impulses in short time) so (enough) sodium <u>ions</u> enter to reach threshold</p> <p><b>OR</b></p> <p>(Several/repeated impulses in short time) increases membrane potential to reach threshold;</p> | <p>2</p> <p>(2 x AO2)</p> | <p>2. Accept abbreviations for acetylcholine e.g., ACh, Ach, AChol.</p> <p>2. Accept 'to cause depolarisation' or 'to produce 'action/generator potential' for 'to reach threshold'.</p> <p>2. Accept Na<sup>+</sup> for sodium ions.</p> |

| Question | Marking Guidance                                                                                                                                                                                                                                 | Mark                      | Comments                                                                                                                                                                                                                                                                                                                                              |
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| 07.3     | <p>1. Less/no acetylcholine/neurotransmitter binds to <u>receptor/s</u>;</p> <p>2. Less/no depolarisation</p> <p><b>OR</b></p> <p>Fewer/no action potential(s)</p> <p><b>OR</b></p> <p>Fewer/no sodium <u>ions</u> enter to reach threshold;</p> | <p>2</p> <p>(2 x AO2)</p> | <p>1. Accept abbreviations for acetylcholine e.g., ACh, Ach, AChol</p> <p>1. Ignore competitive inhibition but reject "active site".</p> <p>2. Accept 'takes longer for depolarisation or action/generator potential to be produced'.</p> <p>2. Ignore 'weaker action potential/depolarisation'.</p> <p>2. Accept Na<sup>+</sup> for sodium ions.</p> |

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                 | Mark                               | Comments                                                                                                                                                                                                                                                                                                             |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07.4     | <p>1. Less/no acetylcholine/neurotransmitter broken down;</p> <p>2. (More) acetylcholine attaches to <u>receptors</u>;</p> <p>3. Depolarisation (of sarcolemma) occurs</p> <p><b>OR</b></p> <p>Action potential(s) produced</p> <p><b>OR</b></p> <p>(Enough) sodium <u>ions</u> enter to reach threshold</p> <p><b>OR</b></p> <p>Fewer/no antibodies attach;</p> | <p>3</p> <p>(1 x AO1, 2 x AO2)</p> | <p>1. Accept (more) acetylcholine/neurotransmitter present/remains.</p> <p>1 and 2. Accept acetylcholine/neurotransmitter remains attached to receptors (for longer) = 2 marks.</p> <p>1 and 2. Accept ACh or other abbreviations e.g. AChol for acetylcholine.</p> <p>3. Accept Na<sup>+</sup> for sodium ions.</p> |